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# Friction and wear performance of polyamide 6 and graphite and wax polyamide 6 composites under dry sliding conditions

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ABSTRACT

The friction and wear performance of pure polyamide-6 (PA-6), 5 wt%, 15 wt% graphite filled polyamide-6 and 4 wt% wax filled polyamide-6 (PA-6+4% wax) sliding against stainless steel under dry sliding conditions were studied using a pin-on-disc tribometer. The influences of filler type, content, applied load and sliding speed on tribological properties were investigated. Tests were carried out at sliding speeds of 0.4, 0.8, and 1.6 m/s and applied load values of 50, 75 and 100 N. Scanning electron microscopy (SEM) was utilised to examine worn surfaces microstructure and wear mechanisms. The results showed that, the friction coefficient for PA6 and PA6 composites increases with the increasing load and sliding speed values while wax filled composite showed insensitivity to the change in load values. In addition, it was observed that their wear rate is highly influenced by sliding speed and insignificantly influenced by the change in applied load values. For the range of materials in this investigation, the lowest wear rate is $6.910^{-15}$ $m^2$/N for 4 wt% wax filled PA-6 composite and the highest rate is $2.12 \times 10^{-14}$ $m^2$/N for pure polyamide-6. The wear mechanism includes transferred film and involves deformation and adhesive wear processes.



## 1. Introduction

Polymer and polymer composites have been increasingly used in various industrial applications such as, aerospace, automotive and chemical industries. This is because these materials provide high strength/weight ratio in comparison to classic material and self-lubricant conditions. However, applications of classic polymer materials are limited due to their low mechanical, thermal and tribological properties. Therefore reinforcements are used to improve their properties. The used reinforcement and additive materials are glass fibre [1–3], $CaCO_3$, [4,5] kaolin, [6,7] talc [8,9], wollastonite [10], and mica [11,12] fillers and MoS2, graphite, carbon, wax and polytetrafluoroethylene (PTFE) and solid lubricants [13,14]. Solid lubricants were proved to be generally helpful in developing a continuous transfer film between the two counterparts and accordingly reduce the frictional coefficient [15].

PA6 is a high performance engineering plastic used in electrical/electronics, automobile, packaging, textiles and consumer applications because of its excellent mechanical properties [6,16–18]. However, limitations in mechanical properties level, the low heat deflection temperature, high water absorption and dimensional instability of pure PA6 have prevented its wide range of applications. Hence, solid lubricants were added to polymide 6 to widen and to increase its application range. Graphite, carbon and wax filled PA6 have found their application in field high velocity applications, without the need for exterior lubrication [13,14].

The friction between polymers can be attributed to two main mechanisms, deformation and adhesion. In this case, the deformation mechanism involves complete dissipation of energy in the contact area while the adhesion component is responsible for the friction of polymer and is a result of breaking of weak bonding forces between polymer chains in the bulk of the material. Some scientists [19–22] reported that the coefficient of friction can, generally, be reduced and the wear resistance improved when the polymers are reinforced with glass, carbon and aramid fibres. In fact, the tribological behaviour of polymers is affected by environmental, operating conditions and by the type, size, amount, shape and orientation of the fibres [23]. A relationship between the wear of the polymers composite content filler and operating parameters is desirable to obtain a better understanding of the wear behaviour. There have been numerous investigations exploring the influence of filler content, test conditions, contact geometry and environment on the friction and wear behaviour of polymers. Franke et al. [24] and Hooke et al. [25] report that the friction coefficient can, generally, be reduced and the wear resistance increased by selecting the right material combinations. Zhang et al. [26], Jia et al. [27], Shi et al. [28] and Anderson [29] observed that the friction coefficient of polymers rubbing against

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metals decreases with the increase in load while Unal and Mimaroglu [30,31] and Shangguan and Cheng [32] showed that its value increases with the increase in load. Watanabe [33] and Bahadur and Tabor [34] have reported that the tribological behaviour of polyamide, HDPE and their composites is affected greatly by normal load, sliding velocity and temperature.

In order to study the influence of graphite, graphite content and wax on tribological properties of polyamide-6, PA-6 composites were produced by twin-screw extruder. Materials contents were PA6+5% G, PA6+15% graphite and PA6+4% wax. Friction and wear tests vs. AISI 440C stainless steel disc were carried out on a pin-on-disc arrangement at dry conditions. Influence of filler type, content, load (range from 50 to 100 N) and sliding speed (range 0.4–1.6 m/s) was explored and evaluated.

## 2. Experimental details

For this study, PA6, graphite and wax materials were supplied from manufacturers. The matrix material polyamide 6 (Domamid) with 1.13 g $cm^3$ density and melt flow index (2.16 kg, 230 °C) 4 g per 10 min was obtained from Domopolymers in Belgium. The graphite was supplied by Sovitec and the wax was supplied by Omya Mining, Egypt. In sample preparation process, the graphite and wax were added to polyamide-6 with 4 wt%, 5 wt% and 15 wt% ratios. The composite granules were prepared using twin screw extruder NR II-75 at temperatures ranging from 220 to 255 °C. The specimens of polyamide-6 and its composites were moulded by using an injection moudling machine at temperatures from 220 to 250 °C. Materials were tested and their physical and mechanical properties were recorded, see Table 1. The friction and wear tests were conducted on a pin-on-disc tribometer. Fig. 1 provides the schematic diagram of the tribometer used. In this test, pin size is Ø6 mm × 50 mm and A4140 stainless steel disc was machined and prepared in Ø100 mm × 5 mm size dimensions with an average surface roughness between 0.31 and 0.36 μm *Ra*. Before each test, PA-6 and its composite pin materials were fixed on a tribometer and rubbed against a metallographic abrasive paper placed on the rotating disc. This pre-rubbing process ensured a full contact of the pin and disc surfaces. Furthermore before each test all samples were ultrasonically cleaned in acetone and then dried. The friction and wear tests were performed at room temperature in an ambient atmosphere. Applied loads ranged from 50 N to 100 N, the sliding speeds ranged from 0.4 m/s to 1.6 m/s, and sliding distance was 4000 m, see Table 2. During the test, friction force was measured by a load cell sensor mounted on the loading arm. The friction force readings were taken as the average of 10 readings every one second during testing time. The coefficient of friction values ($\mu$) were directly obtained from the equipment that records the $\mu$ value by using the following relation:

$$\mu = \frac{Fs}{Fn}$$

where *Fs* is the frictional force and *Fn* is the applied load on the sample. Generally, the specific wear rate is defined as the wear volume normalised by the normal load and the sliding distance. The mass loss ($m$) was measured after each set of run and volume loss ($V$); $V = m/\rho$ was found by using density ($\rho$) of the sample. The following relationship was used to estimate the specific wear rate ($W_{sp}$, $m^3$/Nm):

$$W_{sp} = \frac{V}{FnxL}$$

where $L$ is the sliding distance. Sliding wear data reported here is the average of at least three runs. Before and after each test, the samples were ultrasonically cleaned, dried in warm air and the worn surfaces were characterised by scanning electron microscopy (SEM) (Model Camscan S4) in order to identify the wear mechanism at the surface.

**Table 1**
Polymer materials tested and their physical and mechanical properties.

| Properties | Materials | | | |
|---|---|---|---|---|
| | PA-6 | PA-6+5% Graphite | PA-6+15% Graphite | PA-6+4% wax |
| Density (g/$cm^3$) | 1.13 | 1.16 | 1.21 | 1.11 |
| Tensile strength at break (MPa) | 48 | 66 | 74.1 | 52 |
| Tensile Modulus (MPa) | 2820 | 3400 | 4607 | 2400 |
| Tensile strain at break (%) | 7.09 | 4.23 | 3.71 | 16 |
| Izod impact strength (kJ/$m^2$) | 7.8 | 9.2 | 7.5 | 9.0 |
| Hardness, Shore D | 75 ± 1 | 78 ± 1 | 78 ± 1 | 75 ± 1 |

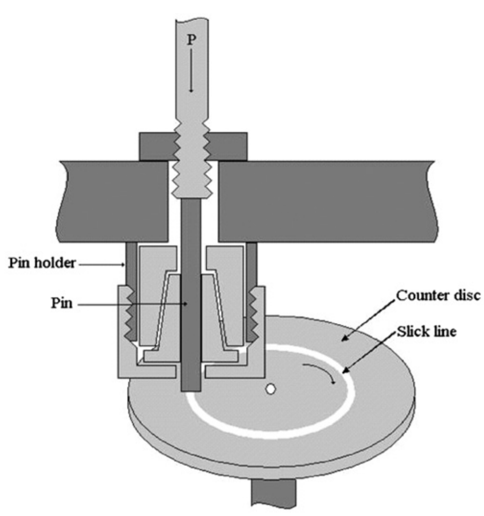


**Fig. 1.** Schematic diagram of the friction and wear test apparatus.

## 3. Results and discussions

Figs. 2 and 3 present the variation of friction coefficient of pure PA-6, PA-6+5% G, PA-6+15% G and PA-6+4% wax composite with applied load at 0.4 m/s and 1.6 m/s sliding speeds respectively. For pure PA-6 polymer, PA-6+5% G and PA-6+15% G composites, the coefficient of friction showed sensitivity to the change in load at load values above 75 N with increasing trend to the increasing load

**Table 2**
Test conditions of the samples.

| Materials | Test temperature (°C) | Sliding distance (m) | Humidity (%) | Applied load (N) | Sliding speed (m $s^{-1}$) |
|---|---|---|---|---|---|
| PA-6<br>PA-6+5% Graphite<br>PA-6+15% Graphite<br>PA-6+4% wax | 21 ± 2 | 4000 | 50 ± 7 | 50<br>75<br>100 | 0.4<br>0.8<br>1.6 |

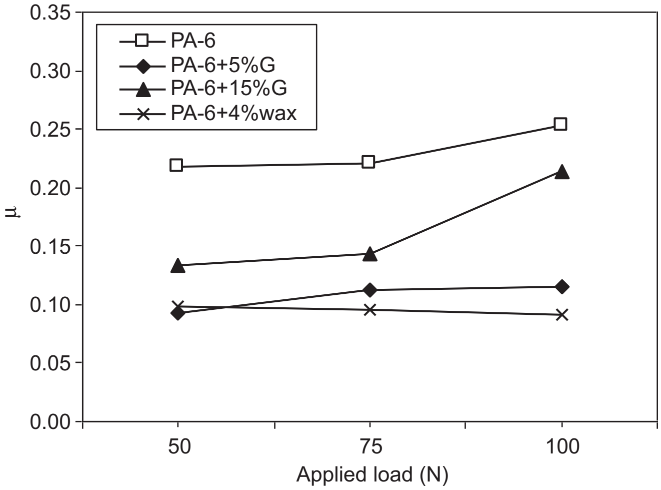


**Fig. 2.** Variation of coefficient of friction with applied load for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (velocity; 0.4 m/s).

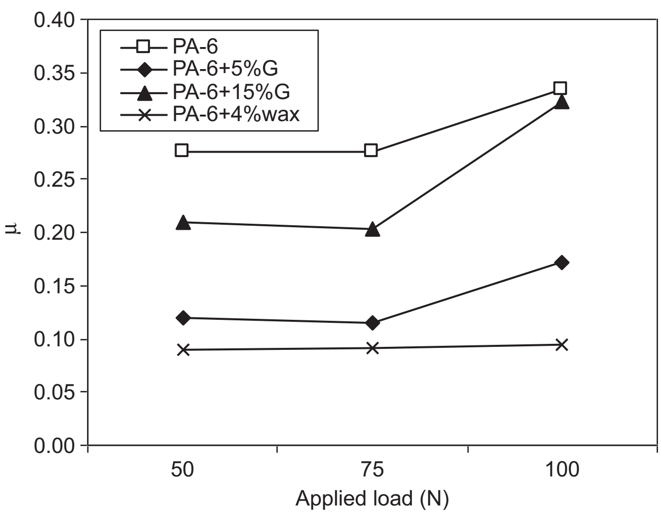


**Fig. 3.** Variation of coefficient of friction with applied load for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (velocity; 1.6 m/s).

values. PA6–wax composite showed insensitivity to the change in load values. Figs. 4 and 5 show the variation of friction coefficient of pure PA-6, PA-6+5% G, PA-6+15% G and PA-6+4% wax composite materials with sliding speed values under 50 N and 100 N applied loads values, respectively. Apart from PA-6+4% wax composite, the coefficient of friction of all tested materials increases with the increase in sliding velocity values. Again PA-6+4% wax composite has shown insensitivity to the change in sliding speed values. As it is known polymers show a visco-elastic behaviour and their deformation under applied load is visco-elastic. So, the variation of friction coefficient with load or pressure follows the equation $\mu = KxL^{(n-1)}$ where $\mu$ is the coefficient of friction, $K$ constant and $n$ is also a constant, its value is $0.66 < n < 1$ and $L$ is the load. According to this equation, the coefficient of friction decreases with increasing load, but when the applied load increases to the limit load values of the polymer, the friction and wear will increase due to the critical surface energy of the polymer material. In addition, this is explained as the frictional heat raised the temperature of the friction surfaces, see Figs. 6 and 7. It is clear from these figures that the pin and disc surface temperature of PA6+5% G and PA6+15% G composites reach about 65 °C which is above the glassy temperature of PA6

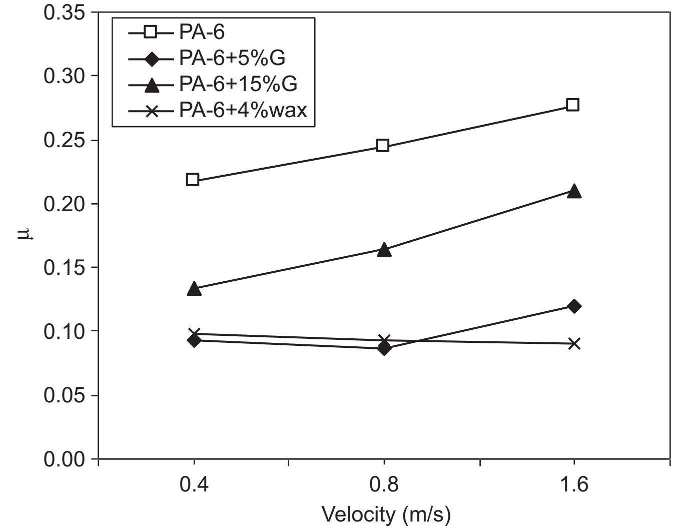


**Fig. 4.** Variation of coefficient of friction with velocity for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (applied load; 50 N).

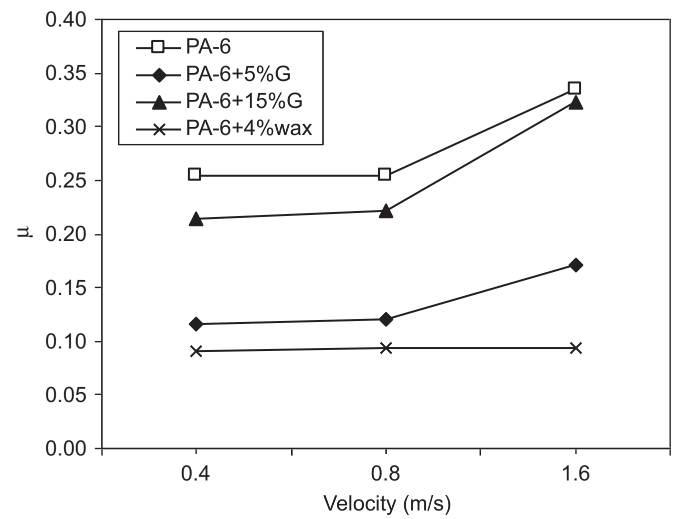


**Fig. 5.** Variation of coefficient of friction with velocity for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (applied load; 100 N).

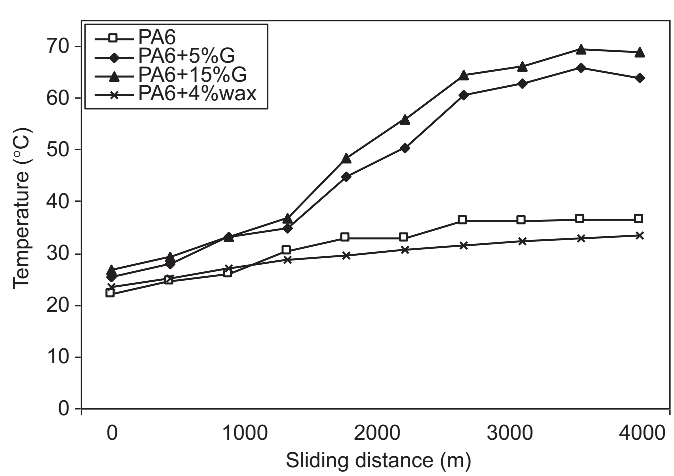


**Fig. 6.** Variation of temperature of pin with sliding distance for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (applied load 100 N; velocity 1.6 m/s).

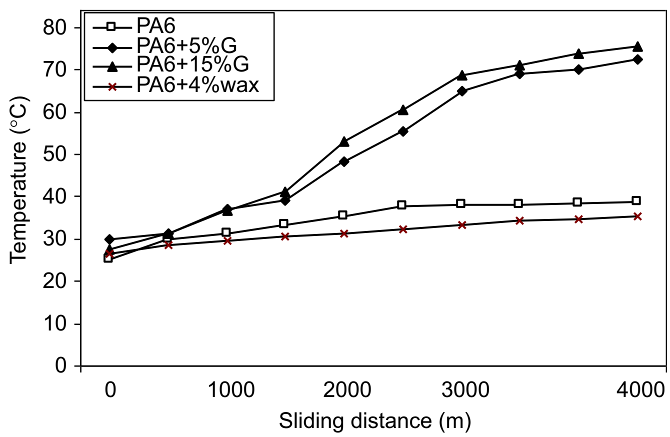


**Fig. 7.** Variation of temperature of stainless steel disc with sliding distance for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (applied load 100 N; velocity 1.6 m/s).

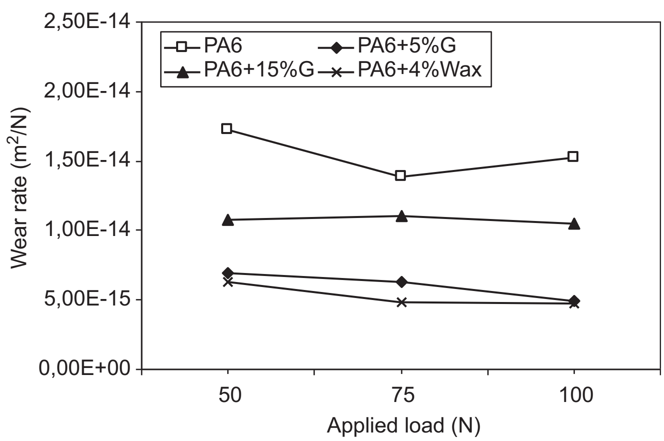


**Fig. 8.** Variation of wear rate with applied load for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (velocity; 0.4 m/s).

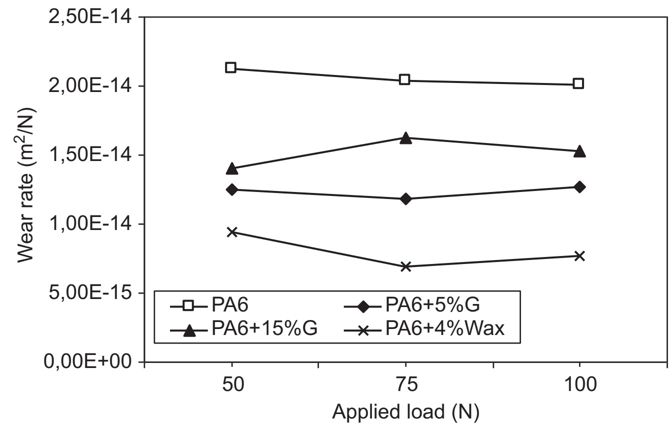


**Fig. 9.** Variation of wear rate with applied load for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (velocity; 1.6 m/s).

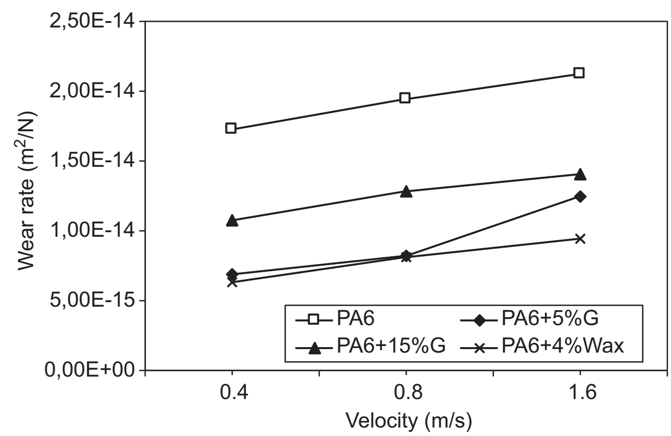


**Fig. 10.** Variation of wear rate with velocity for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (applied load; 50 N).

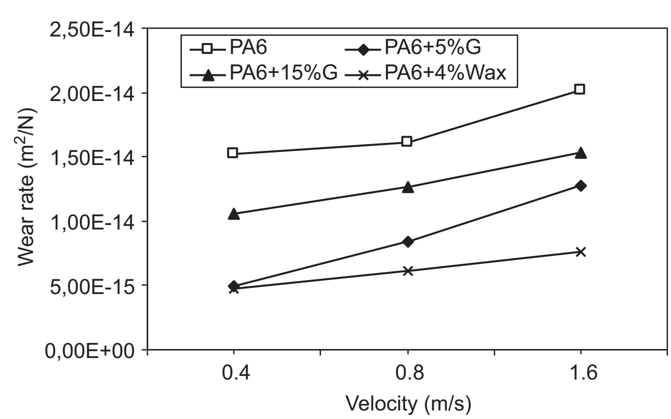


**Fig. 11.** Variation of wear rate with velocity for pure PA-6, PA-6+5% G, PA-6+15% G, PA-6+4% wax composites tested against stainless steel (applied load; 100 N).

($Tg=52$ °C). Hence this resulted in relaxation of polymer molecule chains. In the meantime, molecules of polymer surfaces were pressed, drawn and sheared. Highly active radicals could react with unbroken chains, giving rise to a series of new chain and this is highly active in the presence of graphite filler.

Figs. 8 and 9 show the variation of wear rate versus applied load for pure PA-6 polymer, PA-6+5% G, PA-6+15% G and PA-6+4% wax composites. It is clear from these figures that the wear rate for all tested materials is fluctuating and almost slightly decreases with the increment of applied load values. As it is known wear processes involve fracture, tribo-chemical effects and plastic flow. Transitions between regions dominated by each of these commonly give rise to changes in wear rate with applied pressure. So, this result is closely related to structure characteristics, and chemical effects occurred in frictional processes as well as transfer film formation on the counter-face disc material. Moreover, Figs. 10 and 11 present the variation of wear rate with sliding velocity under 50 and 100 N load values respectively. It is clear from these figures that the wear rate values increase with the increase in sliding speed values. In summary the friction and wear of tested materials are much sensitive to change in speed rather than to the change in applied load values. The addition of graphite and wax resulted in large improvement in lowering friction coefficient and wear rate of PA polymer. It is well known that graphite is a potential candidate which could form a transfer film on the sliding counterpart [25,35]. Furthermore, graphite atoms were arranged in a hexagonal unit cell within each layer [32] and these layers are linked by weak van der Waals bonds, which may be easily broken by shear force under sliding conditions. In general, the abundant filler debris pulled out from the matrix during the sample sliding against the steel counterpart might stick together because of the repeated plastic deformation in the periodic sliding process. So, uniform film was formed on the worn sample surface and reduces the adhesion between the materials and counterparts, which led to decrease in wear rate and in friction coefficient [36,37]. A gain wax form a more stable and active transfer film on the counterface disc which led to a higher drop in wear rate of the material. It is seen that among tested polyamide and polyamide composite materials, the lowest wear rate is for PA-6+4% wax with a value of $6.9\times10^{-15}$ $m^2/N$ followed by PA-6+5% G with $1.0\times10^{-14}$ $m^2/N$ and PA-6+15% G with $1.4\times10^{-14}$ $m^2/N$ and the highest value is $2.12\times10^{-14}$ $m^2/N$ for pure PA6. This means that wax presence in the composite presents 200 times lower wear rate of material. On the other hand, Table 1 shows that addition of graphite led to 54% increase in tensile strength to break, 63% increase in *E*-value, 48% drop in strain to break and 18% increase in impact resistance while adding wax led to insignificant change in strength to break, 15% drop in *E*-value and 125% increase in strain to break and 15% increase in impact resistance. These results and tribology tests results point out the advantage of the addition of wax from tribological point of view with insignificant loss in mechanical properties. Furthermore commercially, wax has the advantage of being cheaper than graphite filler.

Fig. 12 presents the scanning electron micrographs of the pin and disc worn surfaces of the PA-6, PA-6+5% G, PA-6+15% G and PA-6+4% wax composite materials under 100 N load and 0.4 m/s sliding speed test's conditions. It is clear from Fig. 12a that the pin surface of pure PA6 is smooth and rarely shows the presence of transfer film on the disc surface. While PA6+5% G composite shows shiny graphite particles on the pin and the transfer films formed on stainless steel disc surfaces, see Fig. 12b. Fig. 12 also shows that the transfer film developed in high graphite composite is thick with a rough surface finish which increases the probability of removal of

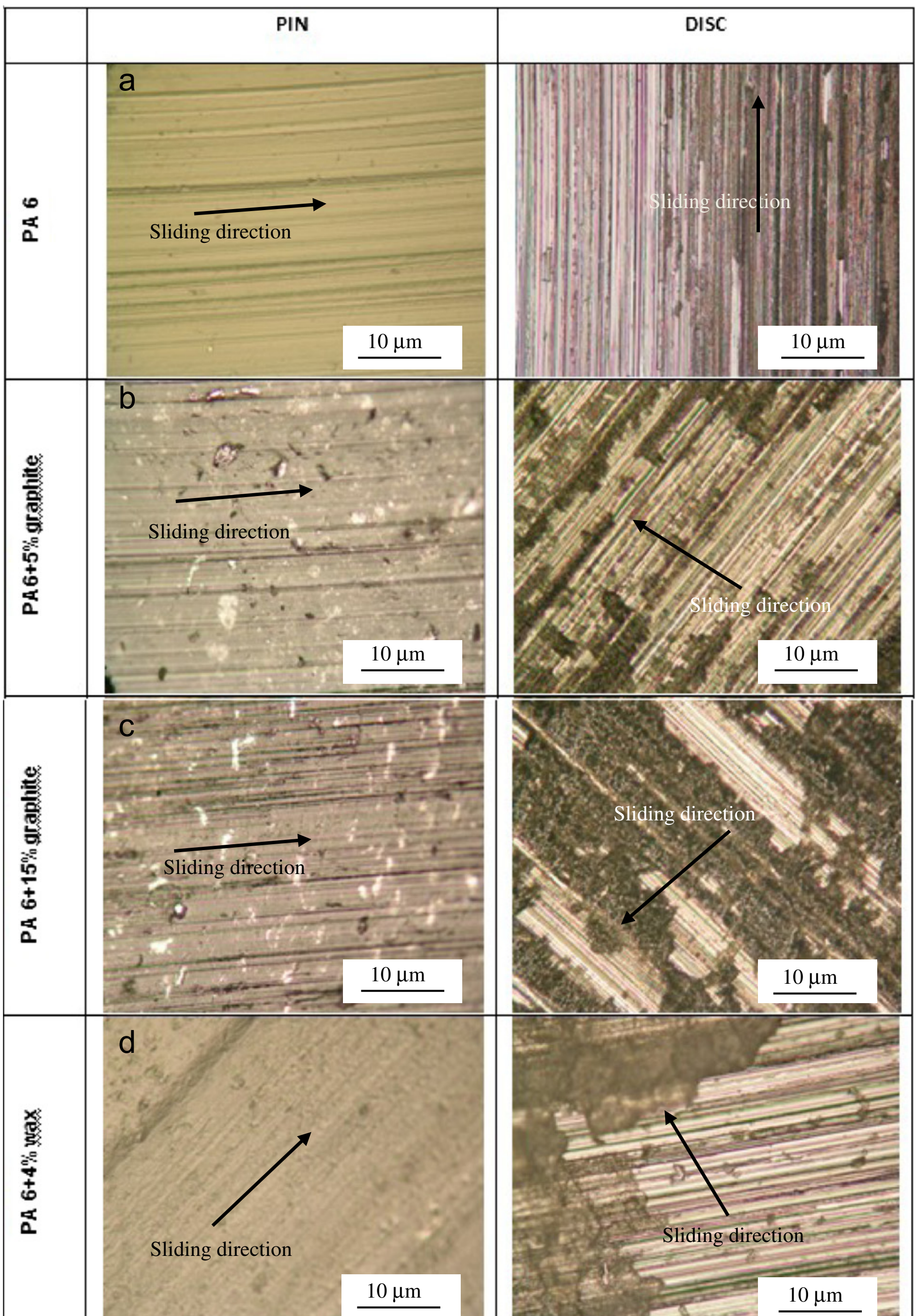


**Fig. 12.** Pin and disc worn surface SEM microscopy at 100 N load and 0.4 m/s sliding speed.

the film during rubbing process hence lead to higher wear rates, see Fig. 12c. In case of wax filler, the formation of transfer film is more active, stable and with different composition. This film protects the rubbing surfaces and has lubricant behaviour.This process lead to much lower friction coefficients and wear rate values, see Fig. 12d.

## 4. Conclusions

The tribological performances of pure PA6 and PA6 composites were investigated and the conclusions reached are as follows. Among the three investigated PA6 composites, the wax composite exhibited a good performance such as low coefficient of friction and high wear resistance. The maximum reduction in wear rate was obtained by PA6+4% wax composite. The specific wear rates for pure PA-6, PA-6+G and PA-6+4% wax composites are in the order of $10^{-14}\,m^2/N$, $10^{-14}\,m^2/N$ and $10^{-15}\,m^2/N$ respectively. The PA6+4% wax composite is 20 and 200 times more resistant under dry sliding conditions as compared to PA6+15% G composite and pure PA6, respectively. For the specific range of applied pressure and sliding speed explored in this experimental study, the sliding speed effect is more on the tribological properties of PA6 and its composites. The wear mechanism shows transfer film development and deformation and adhesive wear processes. Finally, wax being a lot cheaper product than graphite, adding it to PA6 means a high drop in the cost of the PA6 composite material.

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